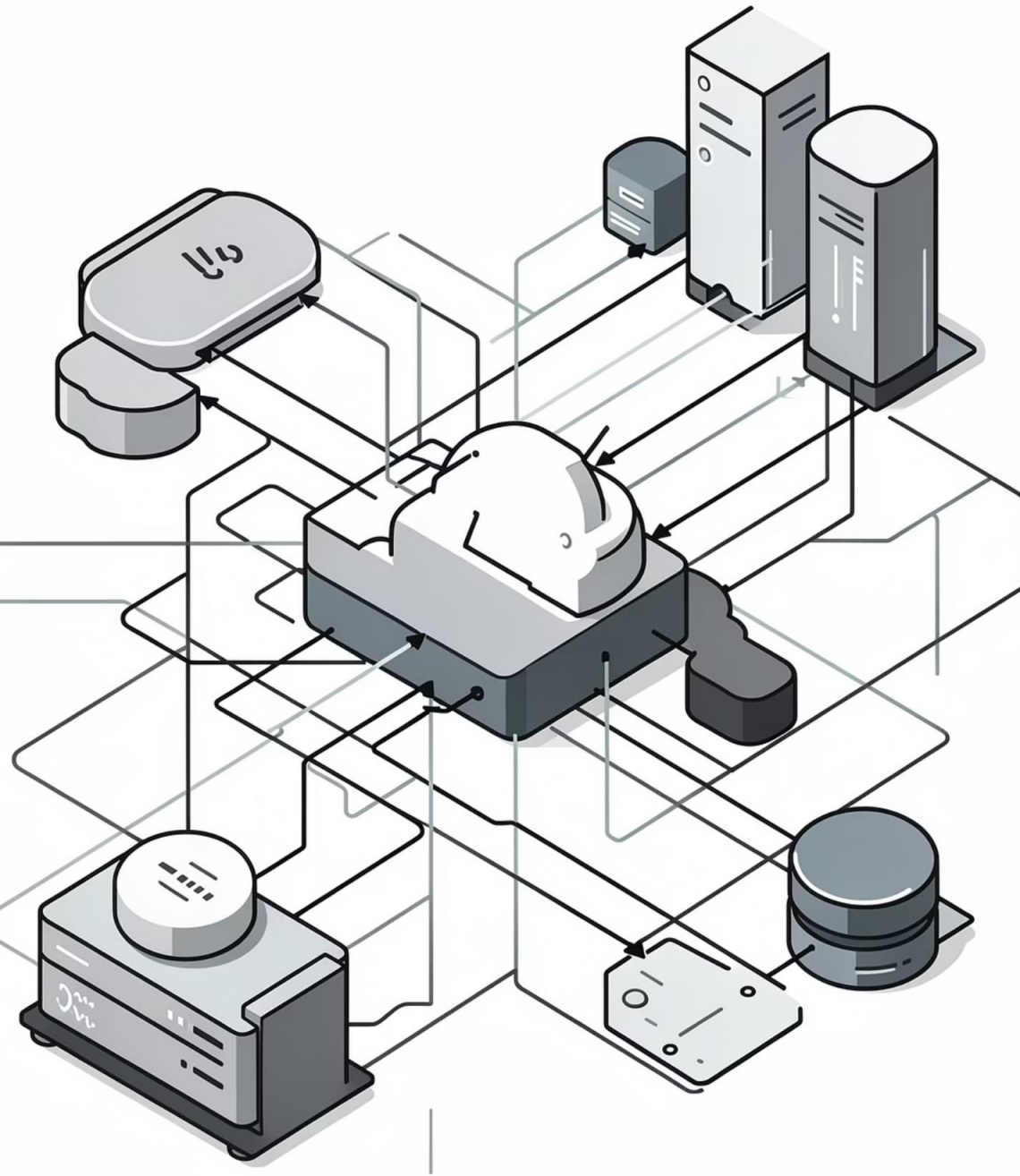




# Monthly Progress Checkpoint

**Sufi Hassan Asim**

January 2026

DevOps & Cloud Learning + Project Onboarding

# Month at a Glance



This month marked a significant transition in my DevOps journey, focusing on two critical areas that form the foundation of effective cloud operations.

By month's end, I successfully transitioned from training environments into understanding and actively supporting production systems.

## Cloud & DevOps Fundamentals

Building strong technical foundation through AWS and automation

## Real Project Onboarding

Transitioning from training to live company systems

WEEK 1

JAN 1 – JAN 7

# Cloud Foundations & Certifications

## AWS Core Concepts

Mastered fundamental cloud architecture principles and AWS service ecosystem

## Hands-On Deployment

Practiced deploying websites on AWS, gaining practical infrastructure experience

## Technical Courses

Completed multiple certification-track courses covering cloud best practices

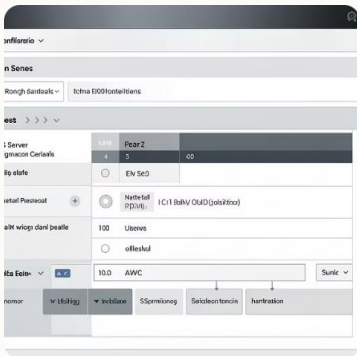
## Linux & CLI Mastery

Strengthened command-line proficiency for cloud operations

## Key Outcome

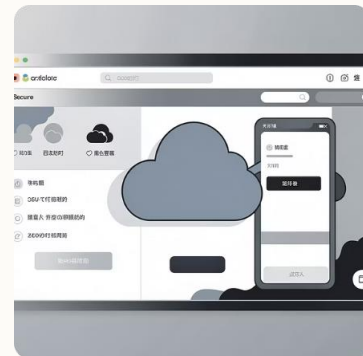
Established a robust technical foundation essential for understanding and managing real-world infrastructure environments.

# Servers, Automation & Secure Access



## EC2 Server Management

Gained hands-on experience configuring and managing cloud servers



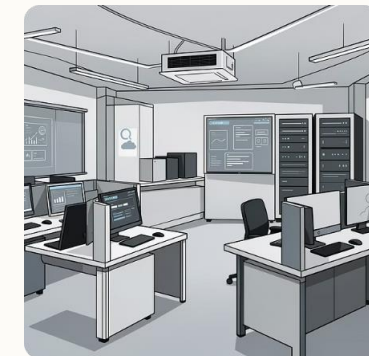
## AWS Systems Manager

Learned secure remote access protocols for production environments



## Infrastructure Templates

Practiced automated infrastructure setup using IaC principles



## Practical AWS Labs

Completed multiple lab exercises reinforcing cloud concepts

 **Key Outcome:** Developed confidence working directly with servers and navigating complex cloud environments.

# CI/CD & Real Architecture Exposure



## Learning Focus

01

### Deployment Pipelines

Learned how automated CI/CD workflows streamline software delivery

02

### Git to Cloud Flow

Understood version control integration with cloud deployment processes

03

### Modern Architecture

Studied three-tier architecture: frontend, backend, and database layers

04

### Production Setups

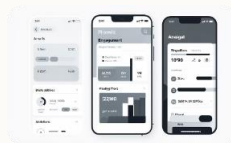
Began working with production-grade configurations and environments



**Key Outcome:** Clear understanding of how modern applications are built, tested, and deployed in cloud environments.

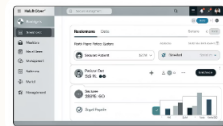
# Real Project Onboarding & Support

This week represented a pivotal transition: moving from training environments into **active company projects**. I was officially onboarded to three production systems, and assigned the 4th project access just 1 day ago, gaining access to their infrastructure and beginning the process of understanding their architectures.



## NGAGE

Reviewed system architecture and deployment workflows



## TBS Health

Gained access to production environment and began understanding health tech infrastructure



## Sindh Ombudsman CMS

Familiarized with government system requirements and compliance standards



## Novavia

Assigned to design and deploy a secret stack for infrastructure

 **Key Outcome:** Now familiar with real project structures, deployment processes, and production environment operations.

# FBR Digital Invoicing Integration



Contributed to a critical **government API integration** project for the Federal Board of Revenue's digital invoicing system. This work required precision and attention to official compliance standards.

## My Contributions

📄 **Result:** Integration testing completed successfully.

- Studied official FBR technical documentation and API specifications thoroughly
- Successfully generated multiple valid test invoices meeting FBR standards
- Performed comprehensive API testing in sandbox environment
- Assisted in the process of obtaining production access token approval



# Production Investigation: TBS Health

Gained valuable exposure to real-world production troubleshooting when the TBS Health application encountered performance issues in the live environment.

- 1

Problem Identification

Application returned **"Too Many Requests (429)"** errors to users
- 2

Investigation

Helped analyze system behavior and identify potential bottlenecks
- 3

Analysis

Reviewed application logs and database connection pool usage patterns
- 4

Learning

Understood how backend resource limits and database connections affect performance

## Key Outcome

This experience provided invaluable insight into how production systems are analyzed, diagnosed, and troubleshooted under real-world conditions. Understanding the relationship between application architecture and performance limitations is crucial for effective DevOps support.

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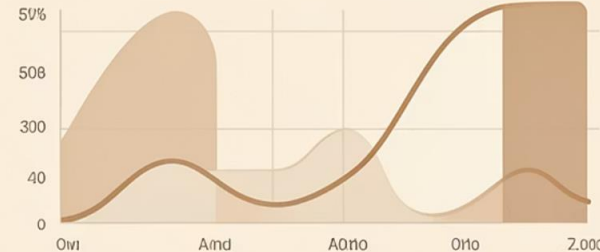
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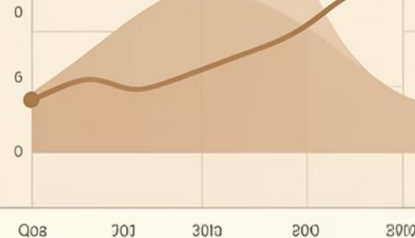
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# Firestore Private Key Secret Management

Implementing a secure and automated solution for managing sensitive Firestore private keys within AWS.



## Define Secrets Stack (CDK)

Create a dedicated CDK stack for the Firestore private key secret.



## Integrate Secrets Stack

Update CDK app entry to include the new Secrets Stack.



## Grant App Permissions

Update App Service Stack to accept secret name and grant read access.



## Deploy & Set Secret Value

Deploy CDK, then populate the secret via AWS Secrets Manager console.



## Update App Service Code

Modify application code to fetch the secret at runtime using AWS SDK.

📄 **Current Status:** CDK stacks are created in the CI/CD repository. Deployment is done and secrets values are added now testing with real secret values is left which was handed to Afifa Akhtar

# Skills Developed This Month

January marked substantial growth across both technical capabilities and professional competencies, establishing a foundation for effective DevOps operations.

## Technical Growth

- 

AWS Cloud Environments

Deeper understanding of cloud architecture, services, and infrastructure management
- 

Server & Deployment

Improved knowledge of server configuration, deployment processes, and automation
- 

CI/CD Pipelines

Practical understanding of how pipelines work in real production projects
- 

API Integration

Hands-on experience testing and validating government API integrations

## Professional Growth

- System Analysis

Reading and understanding existing architectures and codebases effectively
- Team Collaboration

Working alongside team members on active, production-level projects
- Technical Communication

Clearly documenting and communicating technical findings and progress
- Adaptability

Quickly adjusting to new project environments and technical requirements

# Current Status & Next Focus

## Current Status

### 4 Active Projects

Successfully onboarded to  
NGAGE, TBS Health Novavia,  
and Sindh Ombudsman CMS

### Systems Learning

Actively studying each  
project's architecture and  
operational requirements

### Contributing Support

Providing assistance wherever possible while building deeper  
expertise

## Next Focus Areas

01

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### Deployment Contributions

Take more active role in deployments and environment  
management

02

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### Project-Specific DevOps

Master each project's unique DevOps processes and workflows

03

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### Increased Responsibility

Progressively own more DevOps tasks as understanding deepens